Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Agentic AI · LangGraph · Agent Evaluation · RAG · Python · Angular · AWS · Kubernetes · Mortgage & Housing Domain

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PROFESSIONAL SUMMARY

Senior Software Engineer who builds agentic AI systems and the evaluation infrastructure to make them trustworthy in production. AgentProbe is a homegrown agent evaluation harness: LLM-as-judge scoring, CI regression harness so agent quality can't silently degrade, guardrails, production monitoring, precision/recall/F1 across 1,240+ test cases — the exact "evaluation, observability, and guardrail mechanisms" Rocket's platform requires. LeaseI is a production 8-agent LangGraph system with orchestration, tool calling, retrieval-augmented workflows, and durable state — the exact agentic AI capabilities the role describes. Both pair with 7 years of housing and mortgage compliance software at Entrata, where the eligibility and financial pipelines supported the same homeowners Rocket Mortgage serves. Daily user of Claude Code and Cursor as core development tools. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and AI-integrated data pipelines; prompt-driven development and AI coding agents as core workflow tools; debugging production reliability issues in data-driven features.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Built full agentic AI system (LangGraph): orchestration, tool calling, RAG over regulatory corpus, guardrails as hard enforcement logic, and conditional routing. Evaluated LLM model options for routing decisions; used AI coding agents throughout development workflow.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); HuggingFace fine-tuning + inference API deployment; statistical modeling and validation. Workshops to 600+ students across Python and SQL.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Full-stack ownership end-to-end in housing and mortgage-adjacent domain** (Python, C#/.NET, Java, Angular, PostgreSQL, AWS, Kubernetes) — owned 20+ production modules serving 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints. Owned the HUD eligibility and financial compliance pipelines that directly govern housing decisions for the same customers Rocket Mortgage serves. Zero compliance violations. Zero security incidents.
- **Scalable backend and frontend solutions across distributed teams** — designed and implemented APIs, data integrations, and frontend components (Angular); event-driven architecture (RabbitMQ) with idempotency guarantees; multi-tenant PostgreSQL; Docker + Kubernetes; AWS (S3, EC2, RDS, Lambda, Step Functions); 200+ SQL optimizations (30% performance gain, 50% scalability improvement). Collaborated with global engineering teams in India on product delivery.
- **Observability, testing, and team growth** — CI/CD pipelines (GitHub Actions) with automated testing; OpenTelemetry + Prometheus + Grafana (40% debugging time reduction); TDD across all modules; 300+ code reviews; mentored 5+ engineers. Operated in ambiguous environments, translated business needs into reliable technical solutions.

PROJECTS

AgentProbe: Agent Evaluation, Observability & Guardrails LLM-AS-JUDGE CI REGRESSION

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- Homegrown agent evaluation harness: LLM-as-judge classifiers, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), **CI regression harness with golden datasets** so agent quality can't silently degrade in production, multi-provider LLM routing with cost/latency controls, guardrails, prompt injection defense, offline + online evaluation loops, trace-level observability. Built to answer the hardest question in production AI: is this agent actually behaving the way we think it is?

Leasel: Agentic AI System with RAG & Tool Orchestration LANGGRAPH HOUSING COMPLIANCE

Python · LangGraph · FastAPI · Angular-compatible REST APIs · MCP · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production 8-agent LangGraph system: agent orchestration, tool calling, retrieval-augmented generation over housing compliance corpus (hybrid search: TF-IDF + semantic reranking, citation grounding), durable state management, conditional routing, HITL escalation, and prompt architecture governing consistent agent behavior. 93.6% precision and 87.9% F1 across 1,240 housing compliance test cases, evaluated with AgentProbe.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), C#/.NET, TypeScript, JavaScript, Java (Spring Boot), Kotlin, SQL, Bash — Angular, React.js, FastAPI, Django, REST APIs, GraphQL

Agentic AI & Evaluation: LangGraph, LangChain, MCP, Agent Orchestration, Tool Calling, Retrieval-Augmented Generation (RAG), Multi-Agent Workflows, Agent Evaluation Harnesses, LLM-as-Judge, Guardrails, Production Monitoring, Observability, Prompt-Driven Development, AI Coding Agents (Claude Code, Cursor), OpenAI API, Anthropic API, Groq, ChromaDB, Qdrant, Hybrid Search, Semantic Chunking

Backend & Infrastructure: PostgreSQL, MySQL, Redis, RabbitMQ, Apache Kafka, Temporal, Microservices, Distributed Systems, System Design, Multi-Tenant Architecture, Docker, Kubernetes (K8s), AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock), Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, TDD, Automated Testing